Hardware-Conscious Performance Engineering: Cache Hierarchy Optimization, NUMA Topologies, and Hypervisor Resource Scheduling for Memory-Hard HPC Workloads

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Abstract—Executing stochastic, non-linear memory-bound high-performance computing (HPC) workloads within multi-tenant virtualized environments presents severe architectural challenges due to cache thrashing, cross-socket interconnect saturation, translation lookaside buffer (TLB) misses, and hypervisor scheduling contention. This paper presents a systematic, hardware-conscious performance engineering framework evaluated across heterogeneous x86-64 and ARM64 architectures hosted on VMware ESXi 8.0 enterprise hypervisors. By aligning execution thread counts with last-level cache (LLC) capacity, eliminating Simultaneous Multithreading (SMT) execution port contention, partitioning working datasets across Non-Uniform Memory Access (NUMA) domains, deploying 1 GB huge pages to eliminate Extended Page Table (EPT) walks, disabling speculative hardware prefetchers via Model-Specific Registers (MSRs), and enforcing proportional-share hypervisor CPU scheduling, we demonstrate aggregate throughput improvements of 35% to 40% in instructions per cycle (IPC) efficiency. Furthermore, our co-scheduled virtualization methodology recovers 6.1 GHz of host compute capacity per hypervisor node while guaranteeing zero degradation to co-located latency-sensitive production virtual machines.

Index Terms—High-performance computing, memory-hard workloads, cache hierarchy, NUMA topology, TLB optimization, huge pages, Model-Specific Registers, VMware ESXi scheduling.

1 Introduction and Architectural Testbed

Modern scientific simulations, cryptographic state verifications, and graph-analytic kernels increasingly rely on stochastic, non-linear memory-bound high-performance computing (HPC) benchmarks with strict working-set latency constraints. Unlike compute-bound dense linear algebra routines that achieve high arithmetic intensity through predictable spatial locality, stochastic memory-hard workloads execute pseudo-random read-modify-write operations over multi-gigabyte state buffers. Consequently, execution throughput is governed by last-level cache (LLC) residency, memory controller turnaround latency, address translation overhead, and coherence bus traffic rather than peak floating-point operations per second.

When deployed inside multi-tenant virtualized infrastructures such as VMware ESXi 8.0, these workloads encounter compounding virtualization penalties. Two-dimensional page walks across guest and host page tables amplify TLB miss latencies, while blind virtual CPU (vCPU) allocation induces Simultaneous Multithreading (SMT) execution stalls and cross-socket QuickPath Interconnect (QPI) or Ultra Path Interconnect (UPI) bottlenecks.

1.1 Formal Workload Definition

We define the target workload as a Stochastic, Non-Linear Memory-Bound HPC Benchmark operating over a shared read-only state table S of size $M_S \approx 2.3$ GB and a per-thread scratchpad buffer B_i of size $M_B = 2.0$ MB. During each iteration k, an active execution thread i computes a non-linear dependency chain:

$$A_{k+1} = f\left(A_k \oplus B_i[\phi(A_k)] \oplus S[\psi(A_k)]\right) \tag{1}$$

where ϕ and ψ denote uniform pseudo-random indexing functions mapping into the thread-local scratchpad B_i and the global dataset S, respectively, and f represents a mixed integer and vectorizable (AVX2) arithmetic transformation. Because $\phi(A_k)$ exhibits zero spatial or temporal predictability, any eviction of B_i from the L3 cache forces a main-memory round-trip penalty, stalling the core reorder buffer (ROB).

1.2 Heterogeneous Hardware Matrix

To evaluate architectural interactions across server, desktop, and unified-memory paradigms, experiments were conducted across four distinct hardware platforms summarized in Table I.

TABLE 1
Heterogeneous Architectural Evaluation Matrix

Node Type	Processor Architecture	Physical Cores	Total L3 Cache	Me
Node A	2× Xeon E5-2680 v4	28	70 MB	Du
Node B	2× Xeon E5-2620 v2	12	30 MB	Du
Node C	Core i5-8500T	6	9 MB	Sin
Node D	Apple Silicon M1/M2	8	Unified SLC	Un

Node Type A represents a high-density dual-socket enterprise rackmount server based on the Intel Broadwell-EP

microarchitecture (14 physical cores and 35 MB shared L3 cache per socket, 56 logical threads total, quad-channel DDR4 per socket). Node Type B represents a mid-density legacy dual-socket platform using Intel Ivy Bridge-EP (6 physical cores and 15 MB L3 cache per socket, 24 logical threads total). Node Type C is a small-form-factor (SFF) client system based on Intel Coffee Lake (6 physical cores without SMT, 9 MB shared L3 cache). Node Type D serves as an architectural baseline featuring an ARM64 unified memory fabric with high-bandwidth, low-latency access shared directly between compute clusters and system-level cache (SLC).

2 Related Work

Performance engineering for memory-bound workloads has been extensively explored across multi-core server architectures, operating systems, and virtualization layers.

2.1 Cache Hierarchy and SMT Resource Contention

The tension between core concurrency and cache capacity has been a persistent challenge in multi-core systems. Semac et al. [1] demonstrated that oversubscribing shared cache tiers with cache-sensitive worker threads leads to severe eviction thrashing, destroying effective instructions-per-cycle (IPC) metrics. Similarly, studies on Simultaneous Multithreading (SMT) by Koufaty and Marr [2] established that resource contention on shared execution ports and low-level caches (L1d/L2) can degrade thread-level parallelism if logical siblings compete for identical microarchitectural execution units. Our work builds upon these principles by formulating a strict per-thread LLC capacity constraint (2.0 MB) that dictates physical core pinning strategies rather than relying on heuristic task schedulers.

2.2 NUMA Topology and Interconnect Optimization

Non-Uniform Memory Access (NUMA) architectures introduce significant performance variability contingent upon thread-to-memory locality. Drepper [3] provided foundational insights into memory access latencies across local and remote interconnects (QPI/UPI), proving that cross-socket traffic severely degrades bandwidth-bound applications. While prior work by Dashti et al. [4] focused on runtime page migration and kernel-level automatic balancing, our findings indicate that dynamic kernel page-scanning introduces unacceptable jitter for deterministic, stochastic HPC workloads. Consequently, we validate the complete suppression of kernel NUMA balancing coupled with static dual-dataset partitioning to eliminate cross-socket bus saturation.

2.3 Virtualization, Translation Lookaside Buffer (TLB), and MSR Tuning

Virtualization layers introduce substantial address translation overheads due to multi-level page walks. Uhlig et al. [5] detailed the performance impact of Extended Page Tables (EPT) and the criticality of huge page allocations in mitigating TLB miss penalties. While 2 MB huge pages are widely adopted in enterprise Linux deployments, the integration of native 1 GB huge pages ($pdpe1gb$) within multi-tenant hypervisors like VMware ESXi remains underexplored in stochastic memory-hard contexts. Furthermore, low-level hardware optimizations involving Model-Specific Registers (MSRs)—such as prefetcher suppression via MSR_PREFETCH_CONTROL [6]—have traditionally been studied in isolation. This paper bridges the gap by synthesizing CPU prefetcher control, huge page TLB elimination, and proportional-share hypervisor CPU scheduling into a unified, co-designed deployment framework.

3 The 2.0 MB L3 Cache Law and SMT Latency Penalty

3.1 Formulation of the L3 Cache Residency Constraint

For each active thread i to execute at core pipeline speed without main-memory stalls, its entire 2.0 MB scratchpad B_i must remain resident within the shared L3 cache. Let C_{L3} denote the available last-level cache capacity per socket and $M_B = 2.0\,\text{MB}$ denote the strict per-thread working-set footprint. Accounting for instruction cache footprints, page table caches, and hypervisor overhead, the maximum sustainable thread count $T_{\max}$ is governed by the capacity law:

$$T_{\max} = \min\left(N_{\text{phys}}, \left\lfloor \frac{C_{\text{L3}}}{2.0\ \text{MB}} \right\rfloor\right) \tag{2}$$

where N_{phys} is the number of physical execution cores on the processor package. When the active thread count T_{active} exceeds $T_{\max}$, the aggregate scratchpad footprint surpasses C_{L3}, triggering continuous conflict and capacity misses across cache ways.

3.2 Empirical Case Study: Cache Thrashing on Node Type C

Node Type C (Intel Core i5-8500T) contains $N_{\text{phys}} = 6$ physical cores but only $C_{\text{L3}} = 9.0\,\text{MB}$ of shared L3 cache. Applying the capacity law yields:

$$T_{\max} = \left\lfloor \frac{9.0\ \text{MB}}{2.0\ \text{MB}} \right\rfloor = 4\ \text{threads} \tag{3}$$

When a naive scheduler spawns 6 threads to match the 6 physical cores, the aggregate scratchpad demand reaches 12.0 MB, exceeding the 9.0 MB L3 capacity by 33.3%. Empirical profiling confirms that allocating 6 threads degrades total node throughput by 25% compared to a 4-thread allocation. Under the 4-thread configuration, the 8.0 MB aggregate scratchpad leaves 1.0 MB of L3 headroom (2.25 MB effective cache allocation per thread), eliminating cache eviction thrashing and maximizing execution pipeline saturation.

3.3 Elimination of Simultaneous Multithreading (SMT) Contention

On SMT-enabled processors (Nodes A and B), each physical core exposes two logical processors that share the L1 data cache, L2 cache, translation lookaside buffers, and back-end arithmetic logic unit (ALU) and vector execution ports. Although Node Type A provides 70 MB of aggregate L3 cache—theoretically sufficient for $\lfloor 70/2.0 \rfloor = 35$ threads—it possesses only 28 physical cores across its two sockets.

Assigning threads to logical SMT siblings introduces severe structural hazards:

1) Execution Port Serialization: Both logical threads issue micro-operations competing for identical 256-bit AVX2 and integer execution ports, stalling instruction retirement.
2) L1/L2 Cache Pollution: Concurrent stochastic accesses from two sibling threads thrash the 32 KB L1d and 256 KB L2 caches, increasing effective access latency from 4 cycles (L1) to over 40 cycles (L3).

On Node Type A (Dual Xeon E5-2680 v4, 56 logical threads), pinning exactly 28 threads exclusively to physical execution units (14 per socket, stride-allocated across physical core IDs) eliminates the 15% to 20% execution stall penalty observed under logical thread oversubscription. Similarly, on Node Type B (Dual Xeon E5-2620 v2, 12 physical cores, 30 MB total L3), testing 14 threads (which satisfies the 30 MB L3 capacity bound of 15 threads but exceeds the 12 physical cores) forced 2 SMT collisions, measurably degrading per-core IPC. Pinning strictly 12 threads to the 12 physical cores (6 per socket) restored deterministic pipeline execution. By contrast, Node Type D (Apple Silicon M-Series) avoids discrete socket interconnects altogether, leveraging its unified memory architecture to sustain flat access latencies across performance cores without SMT port contention.

4 NUMA Domain Topology and Interconnect Bottleneck Elimination

4.1 Multi-Socket Memory Latency Model

In dual-socket enterprise servers (Nodes A and B), physical memory channels are partitioned across two distinct NUMA domains (Node 0 and Node 1). Let τ_{local} represent the memory access latency from a processor core to its local integrated memory controller (IMC), and let τ_{remote} represent the latency of traversing the cross-socket QPI or UPI coherent interconnect to access the remote socket's DRAM. Empirical latency measurements across the testbed establish:

$$\tau_{\text{local}} \approx 60 \text{ ns}, \quad \tau_{\text{remote}} \approx 140 \text{ ns} \tag{4}$$

When the 2.3 GB shared working dataset S is allocated as a single contiguous virtual memory space interleaved uniformly across both sockets, exactly 50% of all stochastic dataset reads $\psi(A_k)$ must traverse the QPI link. Given a remote access probability $p_{\text{remote}} = 0.5$, the effective average memory access time (AMAT) for dataset reads degrades to:

$$\text{AMAT}_{\text{interleaved}} = (1-p_{\text{remote}})\tau_{\text{local}} + p_{\text{remote}}\tau_{\text{remote}} = 100 \text{ ns} \tag{5}$$

representing a 66.7% latency penalty over pure local DRAM access, alongside severe QPI link bandwidth saturation caused by coherent snoop traffic.

4.2 Case Study: Dual-Dataset Partitioning on Node Type B

On Node Type B (Dual Xeon E5-2620 v2), executing 12 physical threads against a single interleaved 2.3 GB dataset saturated the QPI interconnect, capping aggregate throughput at $1,712$ units/s. To eliminate cross-socket traffic, we re-engineered the memory initialization layer to perform strict NUMA-aware dual-dataset partitioning.

Each NUMA domain independently instantiates a dedicated replica of the 2.3 GB working dataset (S_0 pinned to Node 0 DRAM channels, and S_1 pinned to Node 1 DRAM channels, consuming 4.6 GB total RAM). Threads executing on Socket 0 reference exclusively S_0, while threads on Socket 1 reference exclusively S_1, driving $p_{\text{remote}} \to 0$ and reducing AMAT to $\tau_{\text{local}} \approx 60$ ns. Upon deploying dual-dataset partitioning on Node Type B, sustained throughput surged immediately from $1,712$ units/s to $2,077$ units/s, representing a $+21.3\%$ net performance gain.

4.3 Elimination of Kernel Automatic NUMA Balancing

Even when threads and memory buffers are explicitly bound to local NUMA domains via libnuma policy calls, the Linux kernel's automatic NUMA balancing daemon (numa_balancing) periodically unmaps page table entries (PTEs) to trap subsequent memory references via soft page faults and evaluate page migration heuristics. For a high-throughput HPC workload with deterministic thread-to-socket affinity, this periodic PTE unmapping injects artificial TLB shootdowns, soft fault handling latency, and kernel scheduling jitter. We eliminate this overhead by enforcing static affinity in the kernel runtime configuration:

```
sysctl -w kernel.numa_balancing=0
```

Disabling automatic NUMA balancing guarantees that local PTE mappings remain valid for the duration of the benchmark execution.

5 Memory Virtualization and TLB Hierarchy: 4KB vs. 2MB vs. 1GB HugePages

5.1 Mathematical Analysis of TLB Miss Rates and Page Table Walks

Because every iteration of the workload computes a pseudo-random address $\psi(A_k)$ across the 2.3 GB dataset and $\phi(A_k)$ across the 2.0 MB scratchpad, virtual-to-physical address translation becomes a primary bottleneck. Under a virtualized hypervisor (VMware ESXi 8.0), a TLB miss triggers a two-dimensional hardware page walk across the 4-level Guest Page Table (gPT) and the 4-level Host Extended Page Table (EPT), requiring up to:

$$N_{\text{accesses}} = (4+1) \times 4 + 4 = 24 \text{ memory references} \tag{6}$$

For a per-socket working set of $W_{\text{socket}} \approx 2.35$ GB (2.3 GB dataset plus 14×2.0 MB scratchpads on a 14-core socket), the required number of page table entries E_{TLB} as a function of page size P is given by:

$$E_{\text{TLB}}(P) = \left\lceil \frac{W_{\text{socket}}}{P} \right\rceil \tag{7}$$

Evaluating $E_{\text{TLB}}(P)$ across the three x86-64 architectural page sizes demonstrates why standard paging collapses under stochastic access patterns:

- Standard 4 KB Pages ($P = 4$ KB): Requires $E_{\text{TLB}} = 588,800$ entries per socket. Because hardware L2 data TLBs (STLB) on Broadwell-EP contain only 1,536 entries, the TLB reach covers less than 0.26% of the working set, resulting in a near-100% TLB miss rate on random accesses.
- 2 MB HugePages ($P = 2$ MB): Reduces the requirement to $E_{\text{TLB}} = 1,168$ entries per socket. Provisioning 2,560 pages across a dual-socket node via

sysctl -w vm.nr_hugepages=2560 encapsulates 100% of both NUMA working sets within the hardware STLB capacity, eliminating multi-level page walks.

- 1 GB HugePages ($P = 1$ GB, native pdpe1gb CPU flag): Collapses the translation requirement to exactly $E_{\text{TLB}} = 3$ TLB entries per NUMA node (6 entries total across the dual-socket host).

5.2 Dynamic 1 GB HugePage Provisioning on Dual-Socket Nodes

With 3 entries per socket, the entire 2.35 GB working set resides permanently inside the ultra-low-latency L1 dTLB (which supports 4 dedicated 1 GB page entries per core on Broadwell-EP), reducing address translation overhead to zero cycles. On a 32 GB Dual Xeon node, we dynamically allocate 6x 1 GB pages (3 on NUMA Node 0, 3 on NUMA Node 1) after flushing reclaimable slab and page caches to eliminate physical memory fragmentation:

```
sysctl -w vm.swappiness=0
echo 3 > /proc/sys/vm/drop_caches
echo 3 > /sys/devices/system/node/node0/hugepages/hugepages-1048576kB/nr_hugepages
```

Transitioning from 2 MB pages to NUMA-pinned 1 GB pages yielded an immediate +5.0% IPC uplift and completely eliminated micro-architectural memory access jitter.

6 Low-Level Hardware Registers (MSR) and UEFI Kernel Lockdown

6.1 Hardware Stream Prefetcher Pollution in Stochastic Workloads

Modern x86-64 processors incorporate aggressive hardware prefetchers designed to detect stride and sequential access patterns, preloading subsequent cache lines into the L1 data cache (DCU) and L2/L3 caches before explicit load instructions are issued. On Intel microarchitectures, four hardware prefetch engines operate per core:

1) L2 Hardware Prefetcher (Bit 0): Fetches additional lines into the L2 cache based on spatial locality streams.
2) L2 Adjacent Cache Line Prefetcher (Bit 1): Automatically fetches the paired 64-byte cache line forming a 128-byte aligned sector on every cache miss.
3) DCU Streamer Prefetcher (Bit 2): Monitors L1d loads for ascending or descending address streams.
4) DCU IP Prefetcher (Bit 3): Tracks sequential load history tied to specific instruction pointers (IP).

In a stochastic, non-linear memory-bound benchmark, each memory access $\psi(A_k)$ is statistically independent of $\psi(A_{k-1})$. When the Adjacent Cache Line Prefetcher or L2 Streamer triggers on a random 64-byte dataset read, it speculatively fetches neighboring cache lines that have a near-zero probability of being referenced. This speculative traffic doubles DRAM bus bandwidth consumption, congests the ring interconnect, and prematurely evicts hot scratchpad lines from the tightly constrained 2.0 MB per-thread L3 partition.

6.2 Intel MSR 0x1A4 Manipulation and Kernel Lockdown Remediation

On Intel Broadwell, Ivy Bridge, and Coffee Lake processors, hardware prefetcher state is governed by Model-Specific Register 0x1A4 (MSR_PREFETCH_CONTROL), where setting bits $[3:0]$ to '1' disables the respective prefetch units.

During deployment on virtualized enterprise nodes configured with UEFI firmware, user-space attempts to write 0xF to /dev/cpu/*/msr with root privileges failed with EPERM (Operation not permitted). Kernel audit diagnostics revealed that enabling UEFI Secure Boot automatically transitioned the Linux security subsystem into integrity lockdown mode (Kernel Lockdown = [integrity]). Under integrity lockdown, the kernel blocks direct modifications to hardware MSRs from user space.

To resolve this security architecture conflict, we executed a two-stage remediation:

1) Virtual Firmware Reconfiguration: Disabled UEFI Secure Boot within VMware ESXi virtual machine Boot Options while retaining guest OS filesystem integrity controls.
2) Lockdown State Verification and MSR Execution: Verified that /sys/kernel/security/lockdown reverted to [none], loaded the msr kernel module, and applied register writes across all active physical vCPUs:

```
modprobe msr
wrmsr -a 0x1a4 0xf
```

The pre-start register modification completed in 3 ms across all cores and eliminated speculative cache pollution, stabilizing L3 hit ratios above 99.4%.

7 Hypervisor Multi-Tenancy and Proportional Scheduling (VMware ESXi 8.0)

7.1 The Multi-Tenant Resource Saturation Problem

When high-throughput HPC compute virtual machines colocate with latency-sensitive enterprise workloads on a shared VMware ESXi 8.0 host, unconstrained vector execution can starve production services. Consider an enterprise Dual Xeon E5-2680 v4 host providing an aggregate physical compute capacity of:

$$C_{\text{host}} = 28 \text{ cores} \times 2.40 \text{ GHz} = 67.12 \text{ GHz} \tag{8}$$

Hosting 21 concurrent virtual machines with default scheduling parameters and 18 vCPUs assigned to compute VMs drove host CPU utilization to 95.0% (63.6 GHz consumed), leaving only 3.5 GHz of unreserved headroom.

7.2 Mathematical Resolution: Thread Right-Sizing and Proportional Shares

We resolved hypervisor contention through two complementary scheduling interventions:

1) vCPU Right-Sizing: Reducing compute allocation from 18 vCPUs (44.5 GHz load) to 16 physical-aligned cores (38.4 GHz load) immediately recovered 6.1 GHz of unreserved physical compute capacity.
2) Proportional Resource Allocation (CPU Shares): Let $\mathcal{V}_{\text{prod}}$ denote production VMs (Normal priority, 1000

shares), and $\mathcal{V}_{\text{hpc}}$ denote HPC VMs (Low priority, 500 shares). The guaranteed minimum CPU entitlement E_j during full host contention is:

$$E_j = C_{\text{host}} \times \frac{S_j}{\sum_{i \in \mathcal{V}_{\text{prod}} \cup \mathcal{V}_{\text{hpc}}} S_i} \tag{9}$$

Under zero-contention intervals, the work-conserving scheduler grants $\mathcal{V}_{\text{hpc}}$ 100% of its requested 38.4 GHz clock cycles at peak IPC. During instantaneous production bursts, the scheduler preempts low-share compute vCPUs at microsecond granularity without crashing guest workloads.

8 Empirical Results and Architectural Guidelines

8.1 Quantitative Performance Evaluation

Table II summarizes the cumulative performance impact across tested platforms.

TABLE 2
Performance Benchmark Evaluation

Platform / Config.	Threads	Memory & TLB State	Throughput
Node A (Baseline Unopt.)	56 (SMT)	4KB Interleaved	8,420 units/s
Node A (Tuned Arch.)	28 (Phys)	1GB NUMA + MSR	11,710 units/s
Node B (Single Interl.)	14 (2 SMT)	2MB Interleaved	1,712 units/s
Node B (Dual-NUMA Part.)	12 (Phys)	2MB NUMA + MSR	2,077 units/s

8.2 Formalized Architectural Guidelines

Systems engineers deploying memory-bound HPC workloads on virtualized infrastructures should enforce:

1) Strict LLC-Bound Provisioning: Calculate $T_{\max} = \min(N_{\text{phys}}, \lfloor C_{\text{L3}}/2.0\ \text{MB} \rfloor)$ and pin threads to physical cores.
2) Per-Socket NUMA Replication: Instantiate independent working sets locally and disable kernel automatic NUMA balancing (kernel.numa_balancing=0).
3) 1 GB HugePage Consolidation: Use NUMA-allocated 1 GB huge pages (*pdpe1gb*) to eliminate multi-level page table walks.
4) Prefetcher Suppression: Disable speculative L2/DCU hardware prefetchers by writing 0xF to Intel MSR 0x1A4.
5) Hypervisor Share Isolation: Right-size vCPUs to retain $\geq$ 15% host GHz headroom and assign Low (500) CPU shares.

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